11. Yield of Wheat Exercise 9 (Chapter 11 Review) presented an analysis of variance of the yields of five different varieties of wheat, observed on one plot each at each of six different locations. The data from this randomized block design are listed here:

DATA SET DS1529

	Location					
Variety	**1**	**2**	**3**	**4**	**5**	**6**
A	4 35.3	2 31.0	3 32.7	3 36.8	3 37.2	3 33.1
B	1 30.7	3 32.2	2 31.4	2 31.7	2 35.0	2 32.7
C	5 38.2	4 33.4	4 33.6	4 37.1	4 37.3	5 38.2
D	3 34.9	5 36.1	5 35.2	5 38.3	5 40.2	4 36.0
E	2 32.4	1 28.9	1 29.2	1 30.7	1 33.9	1 32.1

a. Use the Friedman F_r-test to determine whether the data provide sufficient evidence to indicate a difference in the yields for the five different varieties of wheat. Test using $\alpha = .05$.

b. When the analysis of variance was performed in Chapter 11, the ANOVA F-test produced the test statistic $F = 18.61$ with p-value $= .000$. How do these results compare with results of the Friedman F_r-test in part a? Explain.

H_0: The five distributions are the same

H_a: At least one distribution is different.

$T_1 = 4+2+3+3+3+3 = 18$

$T_2 = 1+3+2+2+2+2 = 12$

$T_3 = 5+4+4+4+4+5 = 26$

$T_4 = 3+5+5+5+5+4 = 27$

$T_5 = 2+1+1+1+1+1 = 7$

$$F_{r\,STAT} = \frac{12}{bk(k+1)} \sum T_i^2 - 3b(k+1)$$

$$F_r^* = \frac{12}{6\times5(5+1)}(18^2+12^2+26^2+27^2+7^2) - 3\times6\times(5+1)$$

$$= 20.13$$

$\chi^2_{4,0.05} = 9.49$ RR: $\{F_r \mid F_r > 9.49\}$

$20.13 > 9.49 \Rightarrow$ Reject H_0

There's sufficient evidence to say that at least one distribution is differents.

b. $\chi^2_{4,\,0.01} = 13.28$, $13.28 < 20.13$

$\Rightarrow$ p-value < 0.01 $\Rightarrow$ Reject H_0

$\Rightarrow$ results are the same.